# NoteMates AI App

**THE PORTFOLIO**

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## Introduction

This portfolio documents my journey exploring generative AI as part of the AWS Cloud Practitioner learning plan. Over five parts, I take one idea from a customer problem to a working prototype, using Amazon PartyRock to test it quickly and learn how AWS generative AI services fit together.

My project is ***NoteMates***, a friendly study buddy for students preparing for the AWS Cloud Practitioner exam. Study notes pile up across documents, and finding one answer can take longer than learning it. NoteMates lets students talk to their notes instead. They can add their own notes, use AWS documentation, or combine both, then ask questions in plain English, take practice quizzes, review flashcards, and get quick topic summaries.

*NoteMates* is built in PartyRock, which runs on Amazon Bedrock foundation models, so I could focus on designing the experience, writing effective prompts, and checking the quality and responsibility of the AI's answers. The sections that follow cover the customer problem, the planned features and resources, the first app build, and how I refined it.

*To view the final build of the NoteMates Application, visit:*
https://partyrock.aws/u/kylescritten/KGSyvoxTD/AWS-Cloud-Practitioner-Study-Buddy-NoteMates

## Backend PartyRock App Flow

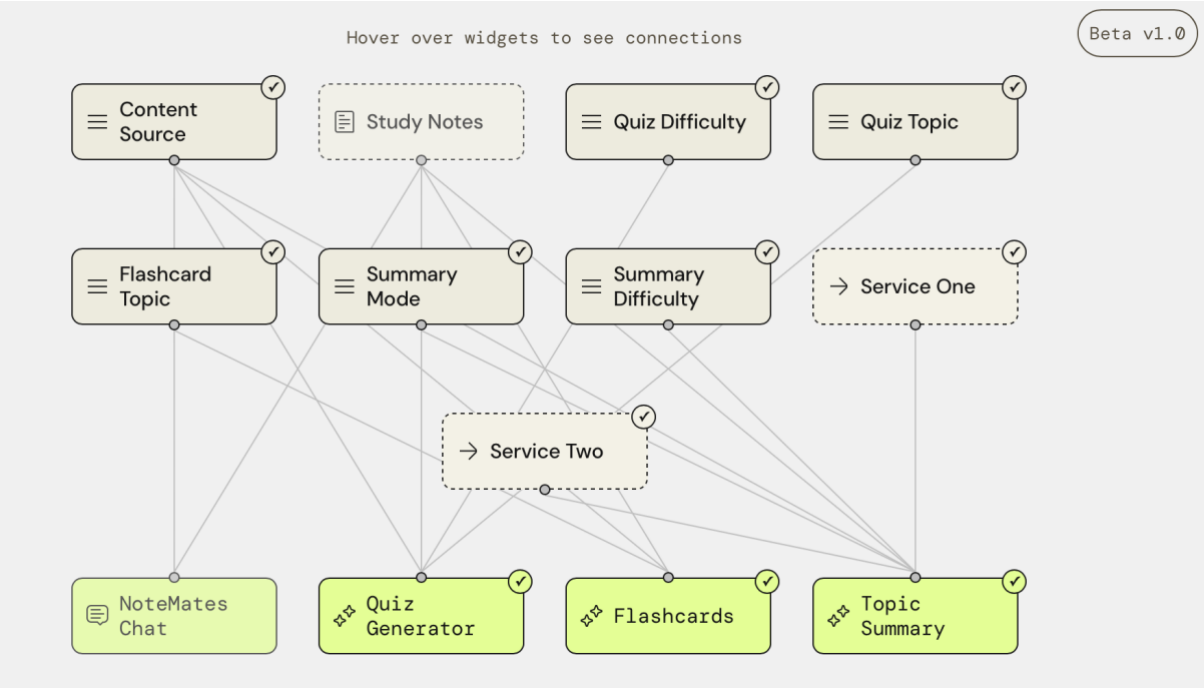



*Figure: How NoteMates widgets are connected.*

## NoteMates Flow Sources

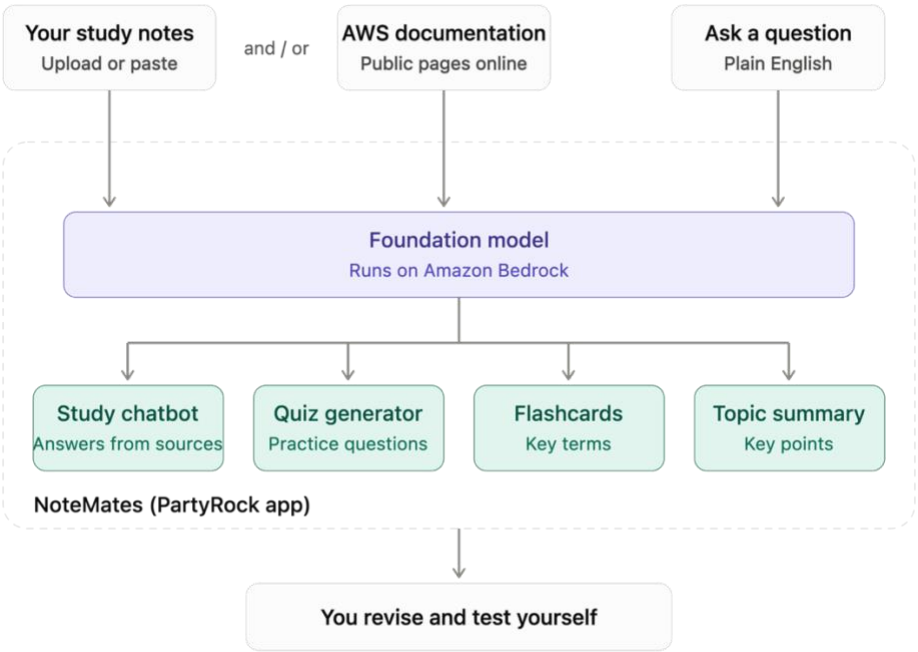



*Figure 1: How NoteMates turns a student's study notes, AWS documentation, or both into answers, quizzes, flashcards and summaries.*

## PartyRock App Architecture

### The Welcome Message

To make NoteMates feel like a study buddy, I wrote a welcome message which will be the first thing the user sees when opening the app:

> "*Hey there, welcome to NoteMates! 👋 Your notes and you make the perfect study pair.*
> *I'm your study buddy for the AWS Cloud Practitioner exam. Here's what I can do:*
>
> - *Answer questions straight from your own notes or AWS documentation*
> - *Quiz you with exam-style practice questions*
> - *Build flashcards for key terms and definitions*
> - *Summarize a topic into quick review points*
>
> *To get started: add your study notes, point me to AWS documentation, or use both, then try asking something like "What's the difference between S3 Standard and S3 Glacier?" or "Quiz me on storage services."*
>
> ***One promise:** I answer from your notes or AWS documentation. If something isn't in them, I'll tell you instead of guessing, and it's always worth double-checking important facts on the official AWS website.*
>
> *Ready when you are. What are we studying today?"*

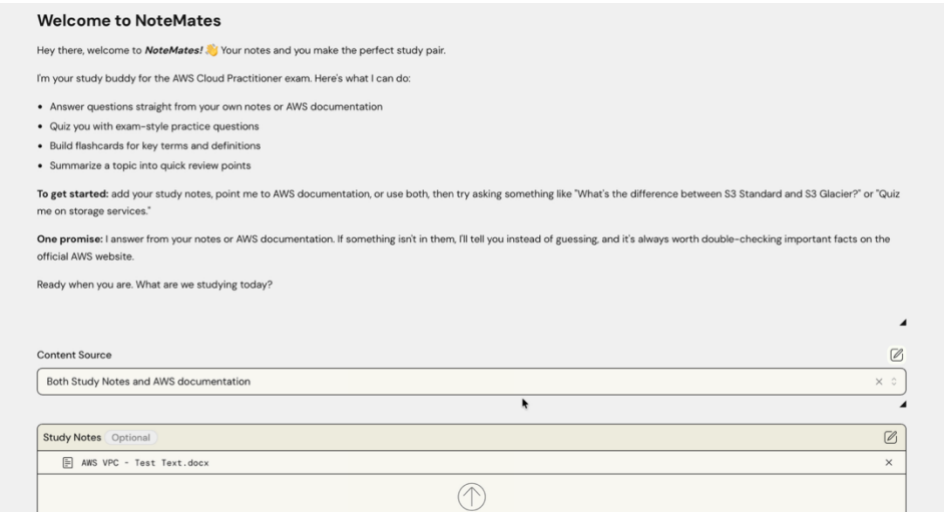



*Figure 2: The NoteMates app in Amazon PartyRock.*

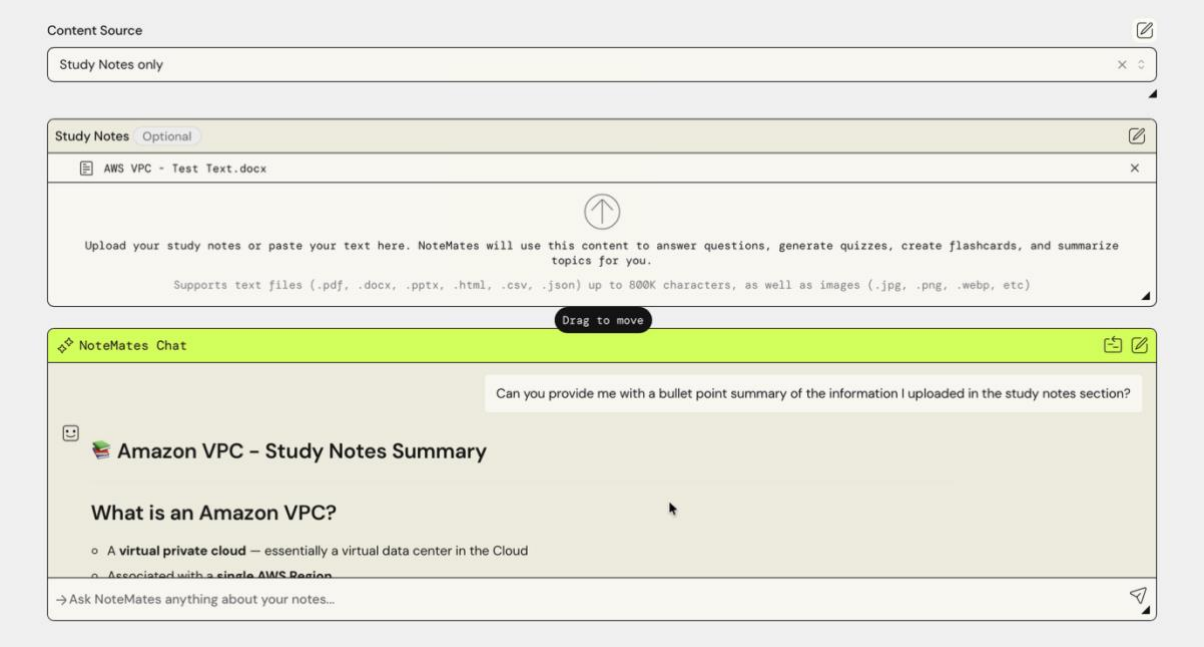
## Content Source And NoteMates Chat Section

### Content Source

A dropdown at the top of the app that lets the student choose what NoteMates draws its answers from. It is currently set to "Study Notes only," so the chatbot and the other AI widgets use just the uploaded document (here, *AWS VPC - Test Text.docx*). The other options let the student switch to AWS documentation or combine both, so the app stays useful even when no notes have been added.

### NoteMates Chat

The chatbot widget where the student talks to their study material. They type a question in plain English into the box at the bottom ("Ask NoteMates anything about your notes..."), and NoteMates replies in the conversation window using the source selected in Content Source. In the screenshot, I asked for a bullet-point summary of the uploaded file, and it returned a formatted summary of the Amazon VPC notes, with a heading, bold key terms, and bullets such as "a virtual private cloud: essentially a virtual data center in the cloud."




**Prompt:** *"Can you provide me with a bullet point summary of the information I uploaded in the study notes section?"*

*Figure 3: The content source and NoteMates chat section*

**Generated Quiz Section**

A set of widgets that turns the selected content source into multiple-choice practice questions. The student picks two settings first: **Quiz Difficulty** (Beginner, Intermediate, or Exam-focused) and **Quiz Topic** (shown here as "Cloud Concepts"). The Quiz Generator then produces 10 questions, each with four options (A to D), a marked correct answer, and a short explanation of why it's right.

The difficulty levels change the style of the questions:

- **Beginner:** simple language and core definitions, with no jargon.
- **Intermediate:** scenario-based questions that focus on how services connect.
- **Exam-focused:** wording that mirrors the real exam.

The Quiz Generator's prompt also has fallback logic: if "Study Notes only" is selected but nothing has been uploaded, it uses AWS documentation instead of stalling.

I tried adjusting the Quiz Generator to show one question at a time, and give a score at the end. So within PartyRock's capabilities I let the NoteMates Chat widget run the quiz interactively. Because the chat already supports back-and-forth conversation, it presents one question, waits for the student's answer, gives feedback, then moves to the next.

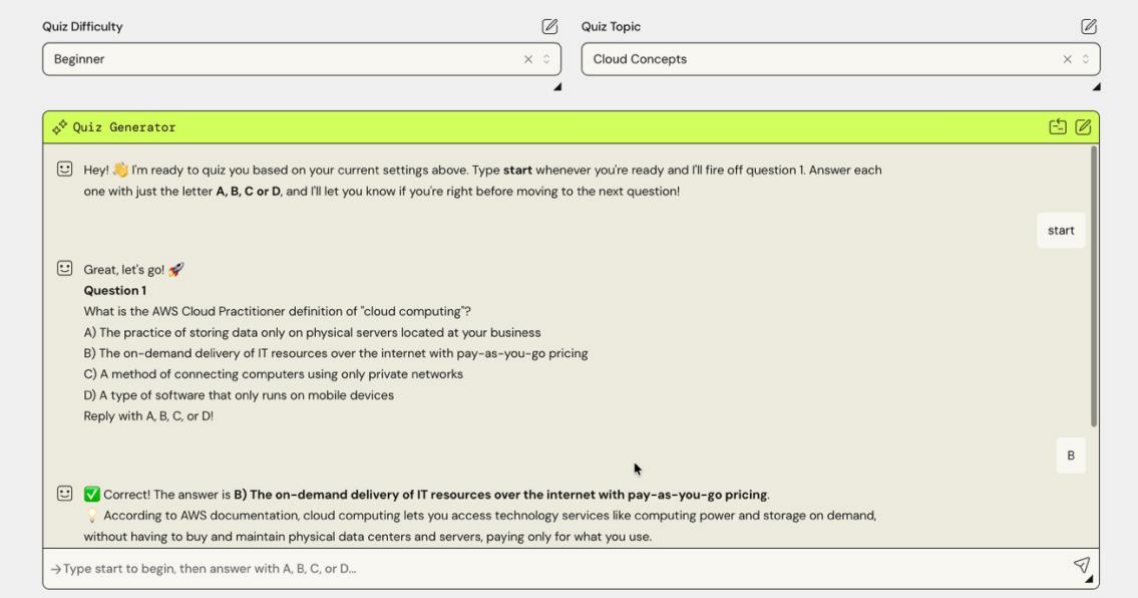



*Figure 4: A generated quiz section*

### Flashcards Section

A widget that pulls out key terms and definitions for quick revision. The student chooses a **Flashcard Topic** from a dropdown (shown here as "Compute (EC2, Lambda)"), and the widget generates a numbered list of terms, each with a short plain-language definition. In the example, it covers Amazon EC2, EC2 instance types, Amazon Machine Images (AMIs), EC2 pricing models, and EC2 Auto Scaling. This gives students a fast way to review one service area at a time instead of rereading their notes.

Students can also select the **Refresh** button in the widget's header and leave the **Flashcard Topic** as it is. NoteMates then generates a new set of key terms and definitions for the same topic, so they can keep revising without repeating the same cards.

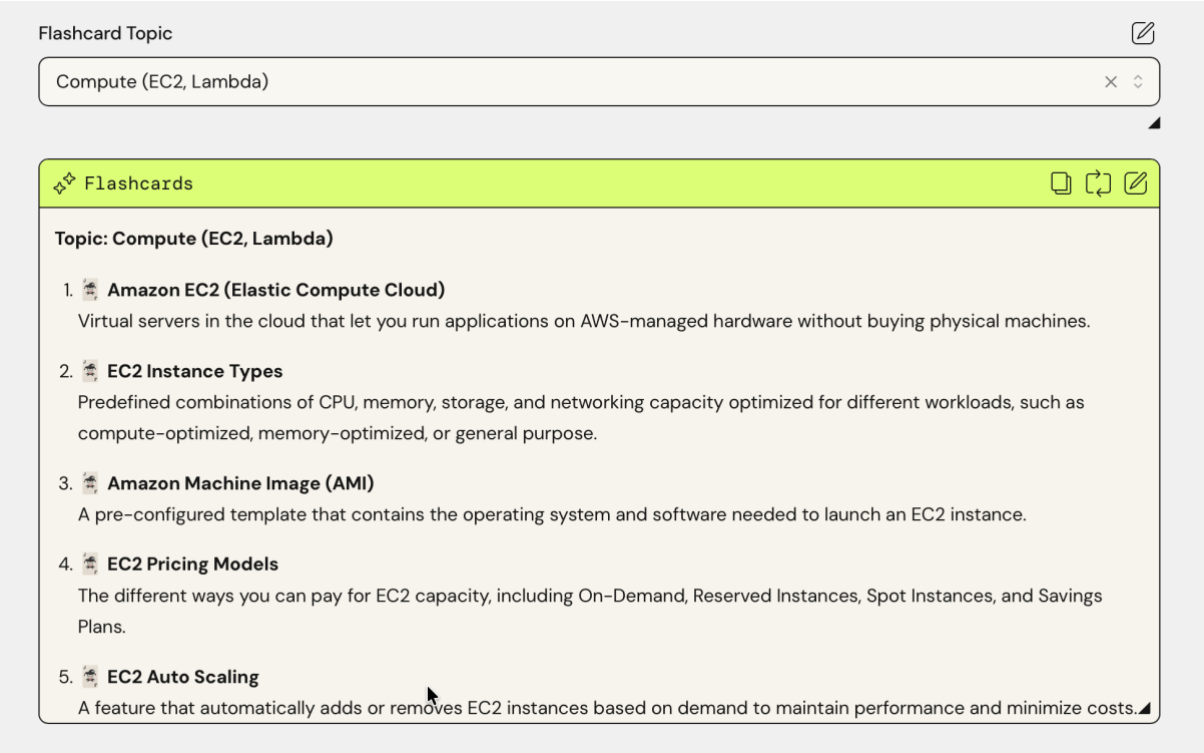



*Figure 5: The flashcards section*

### Topic Summary Section

A widget that condenses the selected content source into quick, focused review material. The student starts with two dropdowns. **Summary Mode** lets them choose

between Quick Review (a short summary of key points), Explain a Service (a breakdown of one AWS service), or Compare Two Services (a side-by-side comparison). **Summary Difficulty** lets them choose between Beginner, Intermediate, or Exam-focused.

Two optional text boxes sit below the dropdowns. **Service One** is used for the Explain and Compare modes (for example S3, EC2, IAM, or Lambda), and **Service Two** is used for Compare mode only (for example RDS or DynamoDB).

The difficulty levels change how the summary is written:

- **Beginner:** simple language and core definitions, with no jargon.
- **Intermediate:** scenario-based, focusing on how services connect.
- **Exam-focused:** wording that mirrors the real exam, with plausible distractors and flags on common traps.

The Topic Summary prompts also have fallback logic: if "Study Notes only" is selected but nothing has been uploaded, they use AWS documentation instead of stalling.

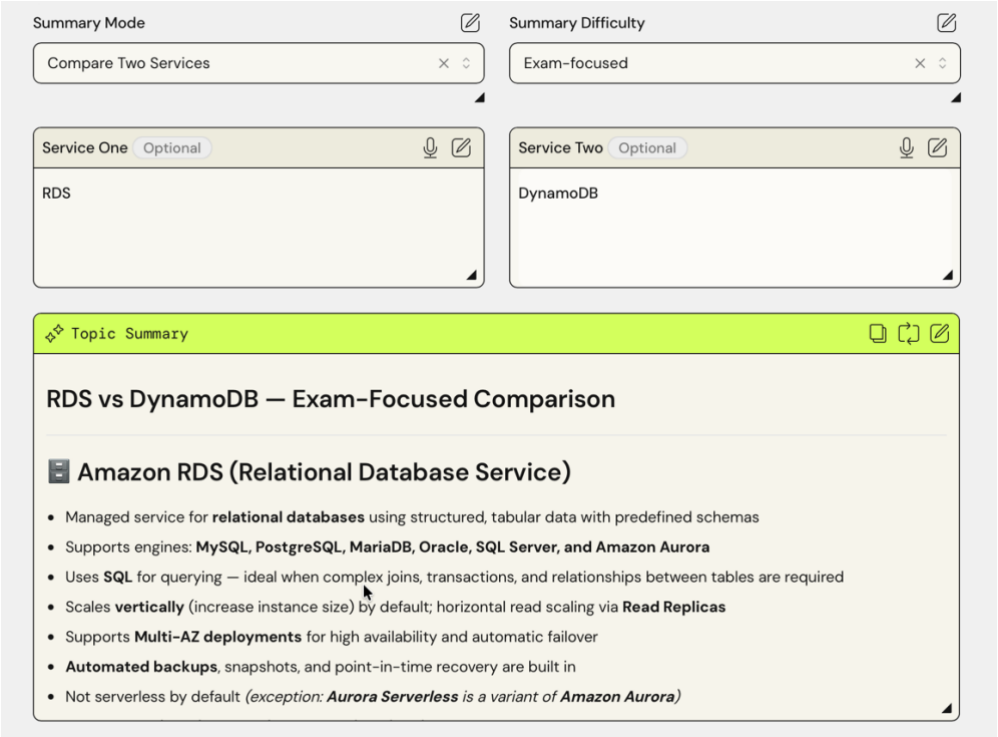



*Figure 6: A topic summary section*

## Prompts used to Generate the Application

- ***Creating the Study chatbot:***
  “You are NoteMates, a friendly study buddy for students preparing for the AWS Cloud Practitioner exam. Check @Study Notes. If it contains real study notes, answer using the notes first, and add AWS documentation knowledge only if the notes don't fully cover the question, saying which part came from where. If @Study Notes says "NO NOTES PROVIDED" or is empty, answer using AWS documentation instead and begin with "No notes added, so I'm using AWS documentation." Keep answers short and clear, in simple language. If you aren't sure of an answer, say so instead of guessing.”

- ***Creating the Quiz generator:***
  “Check @Study Notes. If it contains real study notes, create 5 multiple-choice questions based on them. If it says "NO NOTES PROVIDED" or is empty, create 5 questions based on AWS Cloud Practitioner exam topics from AWS documentation, and begin with "No notes added, so these questions come from AWS documentation." Use exam-style wording, give four options (A-D), mark the correct answer, and add a one-sentence explanation.”

- ***Implementing the Flashcards Feature:***
  “Check @Study Notes. If it contains real study notes, list the 10 most important terms from them. If it says "NO NOTES PROVIDED" or is empty, list 10 key AWS Cloud Practitioner terms from AWS documentation and begin with "No notes added, so these come from AWS documentation." For each term, give a one-sentence definition in plain language.”

- ***Updating Correcting AI Widgets stalling issue across sections:***
  “The app should not wait for @Study Notes to get started in the other widgets. If nothing is uploaded in the @Study Notes widget then AWS Documentation should automatically be used”

- ***Adding a Topic Summary section:***

“Check @Study Notes. If it contains real study notes, summarize them into 5-7 key review points. If it says "NO NOTES PROVIDED" or is empty, ask the user to type a topic, or if none is given, summarize the core AWS Cloud Practitioner exam domains from AWS documentation, beginning with "No notes added, so this summary comes from AWS documentation." Use short bullet points and bold the key AWS service names.”

- ***Updating difficulty across sections:***
  “The difficulty mode selection options are Foundational, Associate, and Professional which is more working-professional focused. This needs to be replaced by Beginner, Intermediate, and Exam-focused across all relevant widgets with a difficulty dropdown because this is aimed at students working towards their professional careers.”

- ***Increasing the number of Quiz questions:***
  “The Generated Quiz is only 5 multiple choice questions, and I would like this to be 10.”

- ***Making the Quiz Generator more interactive:***
  “Change the @Quiz Generator widget to make it more interactive by allowing the user to select an option before displaying the correct answer and one line explanation and moving onto the next question”

## Model Testing: Grounded vs. Direct answers

I tested the *NoteMates* Chatbot in three ways: questions answered from my uploaded notes, the same questions asked of the model directly with no notes and questions not relating to the content covered in my notes.

### Test 1: Question from my notes (Study Notes only)

1. ***Question:*** Does Amazon VPC span across multiple regions?
2. ***Did it match my notes?*** *Yes*
3. ***NoteMates' answer:***

🌐 **Amazon VPC & Regions**

**Based on your study notes — No!** ❌

Your study notes clearly state:

> *"An Amazon VPC lives in **one Region**"*

**Key Points to Remember**

- A VPC is **locked to a single AWS Region**
- However, within that Region, your VPC **can span multiple Availability Zones**
- This is actually how AWS recommends you build for **resiliency and availability**

**Quick Breakdown**

| Resource | Scope |
|---|---|
| **VPC** | **One Region only** |
| **Subnet** | **One Availability Zone only** |
| **Route Table** | Can associate with **multiple subnets** |

💡 **Exam Tip**

This is a **commonly tested concept** on the AWS Cloud Practitioner exam!

Remember:

- **VPC = Regional** 🗺️
- **Subnet = Single AZ** 📍
- If you need resources across Regions, you would need to create a **separate VPC in each Region**

⚠️ Don't confuse **multi-AZ** (within one Region) with **multi-Region** — they are very different concepts!

### Test 2: Same question, no notes (AWS Documentation only)

***Answer:***

**🌐 Amazon VPC & Regions**

**Short Answer: No! ❌**

An Amazon VPC does NOT span across multiple regions. Each Amazon VPC is associated with a single AWS Region only.

**Key Points to Remember**

| Concept | Scope |
|---|---|
| **Amazon VPC** | **Single Region** |
| **Subnet** | **Single Availability Zone** |
| **Default VPC** | **One per Region** |

**How It Works 🏗️**

- When you create an AWS account, separate default VPCs are automatically created in each supported Region.
- If you need resources in multiple regions, you would need to create a separate VPC in each Region
- For added resiliency, a VPC can be divided into multiple subnets across different Availability Zones within that single Region

### 💡 Exam Tip

A common trick question on the AWS Cloud Practitioner exam is confusing **VPC scope (Regional)** vs **Subnet scope (single AZ)**. Remember:

- **VPC = One Region**
- **Subnet = One Availability Zone**

## Test 3: a question my notes don't cover (Study Notes & AWS Documentation)

1. ***Question:*** *"Tell me more about AWS IAM?"*
2. ***What happened:*** The chatbot provided a comprehensive answer from AWS Documentation because the question I asked did not reference information I had in my notes.

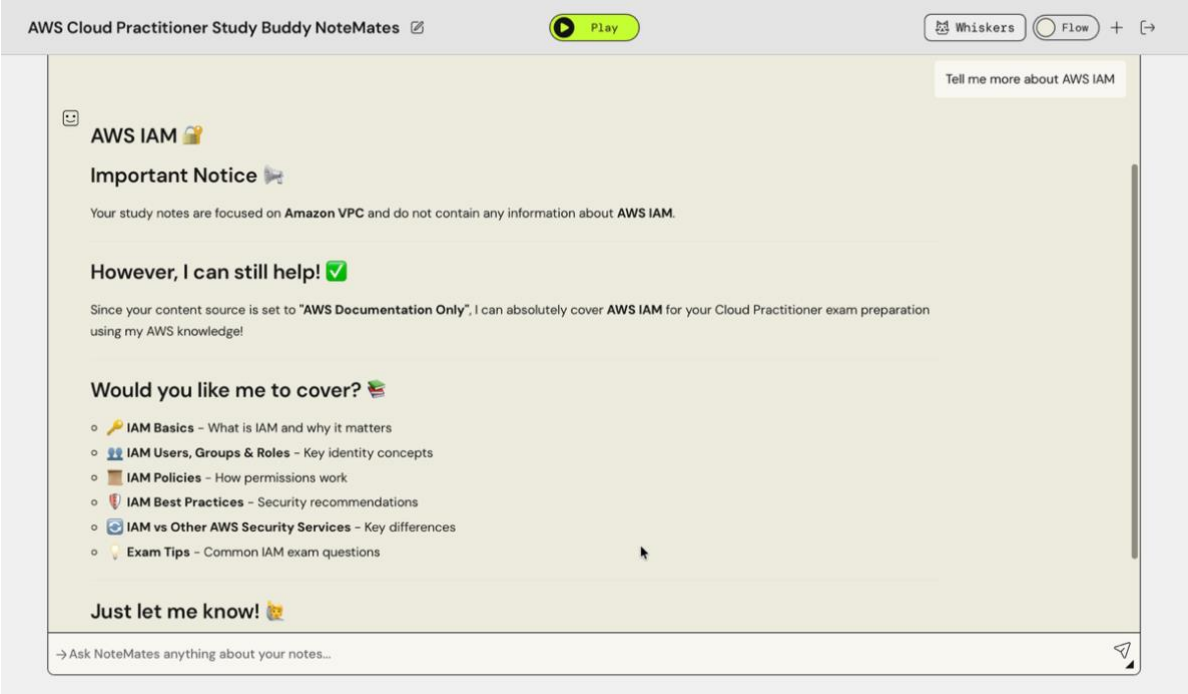



*Figure 7: A Chatbot answer screenshot*

## Challenges and Key Changes

### Issues Identified while building

1. The widgets stall when notes are empty and the source is "Study Notes only". I need to update all AI widgets to automatically fall back to AWS documentation when no notes are provided.
2. Quiz Generator, Flashcards, and Topic Summary widgets all have updated prompts with explicit fallback logic — if "Study Notes only" is selected but nothing is uploaded, they automatically use AWS documentation instead of stalling
3. The @Quiz Generated widget outputs a long list of text. Ideally, it would let users select an answer, show only one question at a time, and move on to the next question only after a selection is made. However, this isn't possible in PartyRock. AI Output widgets display all their generated text at once and can't respond to button clicks or answer selections, so a user's choice can't trigger the next output.
4. The upload section for the User Study Notes does not work, the only way to upload notes is by drag-and-drop the file into the upload field.

### Key Changes after Initial Build

1. The study notes upload has been optional, so that official AWS Documentation is referenced if no notes are uploaded.
2. The welcome/introduction section is more robust and welcoming and includes a subtle “how to” for the app.
3. Quiz Difficulty and Summary Difficulty dropdowns now offer Beginner, Intermediate, and Exam-focused (defaulting to Beginner). Both ***Quiz Generator*** and ***Topic Summary*** prompts now interpret these levels meaningfully:
    a. Beginner → simple language, core definitions, no jargon
    b. Intermediate → scenario-based, focuses on how things connect
    c. Exam-focused → mirrors real exam phrasing, includes plausible distractors and flags common traps
4. The Generated Quiz is only 5 multiple choice questions, and I have changed this to be 10.

5. The Quiz Generator is now a chat-style quiz instead of a one-shot block of text. After clicking start, the student gets one question at a time with options A to D and types their answer letter. Only then does the quiz show whether they were correct, the right answer, and a one-sentence explanation, before moving on to the next question. It still follows the Quiz Difficulty, Quiz Topic, and Content Source settings, and gives a final score after 10 questions. PartyRock doesn't support clickable answer buttons, so "selecting" an option means typing the letter, which is the closest interactive experience the platform allows.

## Conclusion

NoteMates began with a simple frustration: students preparing for the AWS Cloud Practitioner exam collect notes from video lessons, documentation, and practice quizzes, then lose time searching through them for a single answer. I wanted to turn those notes into something a student could talk to. The app is aimed at students working toward professional careers in cloud computing, who need to revise efficiently and test their understanding, not just reread what they've written.

The final app does this through several connected features. A Content Source selector lets the student choose whether answers come from their own study notes or AWS documentation. The NoteMates Chat answers questions in plain English from that source. The Quiz Generator runs an interactive 10-question quiz, one question at a time, with feedback after each answer and a final score. The Flashcards widget produces key terms and definitions by topic, and a Refresh button generates a new set. The Topic Summary offers three modes (Quick Review, Explain a Service, and Compare Two Services). Quiz and summary widgets share three difficulty levels: Beginner, Intermediate, and Exam-focused.

Building it in PartyRock brought challenges and forced some changes to my original plan. My AI widgets stalled when no notes were uploaded, so I rewrote their prompts to fall back to AWS documentation automatically. The first version of the quiz produced all its questions at once, and PartyRock's text widgets can't wait for an answer, so I rebuilt it as a chat-based quiz. PartyRock also has no clickable answer buttons, so students select an option by typing A, B, C, or D. My original "beginner mode" became a difficulty setting across the quiz and summary widgets, and the plan for a database and progress tracking was set aside, which means notes are added fresh each session. Each of these was a trade-off between the ideal experience and what the platform could do, and working within those limits taught me as much as the successes did.

Overall, NoteMates shows how quickly a real student problem can become a working generative AI prototype, and how much the results depend on clear prompts, honest testing, and a willingness to adapt.